

ATHINAI HELLINIKON, Greece

WMO: 167160

Lat: 37.8897N

Lon: 23.7417E

Elev: 43

StdP: 100.81

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.9	3.6	-6.2	2.3	5.8	-4.0	2.7	6.6	10.9	9.1	9.8	9.7	2.6	350	0.383

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.9	35.8	21.2	34.3	21.1	33.2	21.2	24.6	31.5	23.8	30.9	23.1	30.0	4.9	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.2	17.0	29.0	21.6	16.3	28.5	20.8	15.5	27.9	74.8	31.6	71.8	30.9	68.8	30.2	27.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.5	8.4	7.5	-0.3	38.5	2.0	1.3	-1.8	39.4	-3.0	40.2	-4.1	41.0	-5.6	41.9
			-2.1	26.1	1.7	0.8	-3.4	26.7	-4.4	27.2	-5.3	27.6	-6.6	28.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	18.7	10.2	10.7	12.7	15.8	20.8	25.7	28.5	28.6	24.3	19.9	15.5	11.7
	DBStd	7.12	3.15	3.12	2.85	2.53	2.56	2.75	1.85	1.96	2.43	2.80	2.97	3.24
	HDD10.0	96	35	28	9	1	0	0	0	0	0	3	21	
	HDD18.3	1046	251	215	174	83	7	0	0	0	18	93	205	
	CDD10.0	3286	43	47	94	173	333	470	573	577	431	306	166	74
	CDD18.3	1194	0	0	1	6	82	220	314	318	181	65	7	1
	CDH23.3	11898	0	0	2	13	450	2187	3844	3856	1334	208	4	0
	CDH26.7	5015	0	0	0	1	83	873	1822	1838	381	18	0	0
Wind	WSAvg	2.9	2.8	3.0	3.0	2.7	2.6	3.0	3.3	3.3	2.9	2.7	2.6	2.9
Precipitation	PrecAvg	360	47	40	40	25	16	7	7	4	14	44	58	67
	PrecMax	514	177	129	156	113	71	37	61	45	75	192	203	163
	PrecMin	158	0	3	0	0	0	0	0	0	0	0	0	0
	PrecStd	76	37	31	32	25	17	9	15	8	21	42	46	42
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.2	18.6	21.9	25.6	30.9	36.2	37.7	38.1	33.4	28.4	24.0	20.2
		MCWB	14.2	13.7	14.9	16.6	18.5	20.6	21.5	20.9	20.9	20.7	18.3	16.6
	2%	DB	16.9	17.3	19.8	23.1	28.9	34.1	35.6	36.0	31.3	26.9	22.3	18.7
		MCWB	13.5	12.9	14.0	15.2	18.1	20.8	21.3	21.2	20.8	19.5	17.7	15.1
	5%	DB	16.0	16.4	18.5	21.8	27.2	32.6	34.2	34.5	30.1	25.5	21.2	17.7
		MCWB	12.8	12.4	13.3	14.9	18.0	20.6	21.2	21.3	20.6	19.0	17.2	14.4
	10%	DB	15.1	15.5	17.4	20.4	25.9	31.1	33.1	33.2	29.0	24.2	20.1	16.5
		MCWB	12.1	11.9	12.7	14.3	17.7	20.4	21.2	21.3	20.2	18.6	16.4	13.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.1	14.7	15.7	17.6	21.1	24.0	25.3	25.8	24.5	22.5	19.7	17.2
		MCDB	17.6	17.6	20.1	23.2	27.9	31.5	32.2	32.6	30.3	26.6	22.5	20.0
	2%	WB	14.1	13.7	14.7	16.6	20.1	23.0	24.5	24.9	23.2	21.3	18.6	15.8
		MCDB	16.3	16.4	18.7	21.6	26.4	30.5	31.6	31.7	28.5	25.1	21.4	18.1
	5%	WB	13.3	13.0	14.0	15.8	19.2	22.2	23.7	24.0	22.2	20.4	17.7	14.8
		MCDB	15.5	15.8	17.6	20.4	25.2	29.7	31.0	31.1	27.5	24.1	20.7	17.2
	10%	WB	12.3	12.3	13.2	15.0	18.5	21.5	23.0	23.2	21.5	19.5	16.8	13.8
		MCDB	14.8	15.0	16.7	19.4	24.2	28.8	30.2	30.1	27.0	23.2	19.6	16.1
Mean Daily Temperature Range	5% DB	MDBR	6.1	6.8	7.2	7.9	8.8	9.1	9.1	8.9	8.1	7.2	6.4	5.6
		MCDBR	6.6	7.8	8.4	9.1	10.2	10.4	10.4	10.2	9.2	8.2	7.1	6.0
		MCWBR	4.5	5.1	5.1	5.0	4.9	4.6	4.8	4.6	5.1	4.6	4.4	4.2
	5% WB	MCDBR	5.9	7.1	7.6	8.5	9.6	9.6	9.6	9.3	8.4	7.3	6.6	5.4
		MCWBR	4.6	5.1	5.1	5.1	5.2	5.1	5.4	5.6	5.3	4.8	4.6	4.3
Clear-Sky Solar Irradiance	taub	0.350	0.374	0.409	0.453	0.436	0.416	0.420	0.425	0.424	0.396	0.376	0.360	
	taud	2.433	2.349	2.247	2.130	2.242	2.335	2.333	2.309	2.312	2.369	2.392	2.407	
	Ebn at Noon	823	847	848	829	850	865	858	846	824	811	783	780	
	Edh at Noon	86	105	128	152	138	126	125	126	118	102	89	83	
All-Sky Solar Radiation	RadAvg	2.01	2.79	4.18	5.50	6.57	7.50	7.62	6.79	5.20	3.54	2.36	1.75	
	RadStd	0.22	0.33	0.33	0.47	0.32	0.37	0.16	0.14	0.27	0.28	0.23	0.23	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (11 neighbors)	+0.35	N/A	N/A	N/A	N/A	N/A	N/A	-82	N/A	N/A

Nomenclature: See separate page